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CHEMICAL REACTION ENGINEERING (CHE 321)

QUIZ 1

CLOSED BOOK, TIME 15 MIN

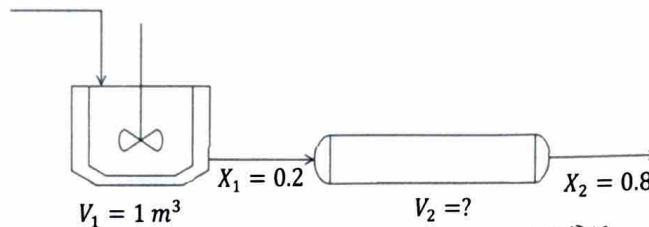
Consider the liquid-phase reaction $A \rightarrow B$, with the rate as a function of conversion given as

$$r_A = -kC_{A0}(1-X) \quad \text{conversion}$$

, where k is a rate constant, C_{A0} is the concentration of species A in the feed.

The reaction is to be carried out isothermally in a continuous stirred tank reactor (CSTR), followed by a plug flow reactor (PFR). Volume and conversion of the CSTR are 1 m^3 and 20%.

Calculate the volume of the PFR required to achieve 80% final conversion.



Hint: The General mass balance equation is $\frac{dN_A}{dt} = F_{A,in} - F_{A,out} + G_A$
 not a function of time

$$X = \frac{F_{A0} - F_A}{F_{A0}}$$

PFR

$$0 = \left[F_{A0} \right]_{in} - \left[F_A \right]_{out} + r_A \Delta V \rightarrow \frac{dF_A}{dV} + r_A$$

then integrate to get:

$$V = \frac{F_{A0} X}{-r_A}$$

$$F_{A0} X = F_{A0} - F_A$$

$$\int_0^{V_1} dV = \frac{F_{A0}}{-r_A} dx$$

$$(V_1 - V_2) = \frac{F_{A0} (X_1 - X_2)}{-kC_{A0}(0.80)}$$

$$1 \text{ m}^3 - V_2 = \frac{F_{A0}}{-kC_{A0}(0.4)} 0.6$$

$$r_A = -kC_{A0}(1 - 0.6)$$

$$r_A = -kC_{A0}(0.4)$$